

A Study of Landslide Susceptibility Mapping using Machine Learning Approach

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Abstract: Natural disasters are a great threat to human life. Landslide is a natural threat to the environment that causes substantial damage to property and loss of life. This affects both the economic and the social aspects. India is one of the world's most vulnerable to landslides. There are various statistical and machine learning methods to predict landslide. Here we have studied critically the various LSM mapping machine learning techniques used in the last three critical years. Various researchers have suggested various ML strategies for LSM. LSM can detect vulnerable landslides that help avoid the occurrence of this natural hazard. In most studies, ML techniques such as SVM, RF and ANN have been studied in this area. Hybrid models also adopted by different researchers for the same analysis and in most of the studies, we found that models were tested using the receiver operating characteristics (ROC)/AUC technique. CNN has also concentrated on LSM in several recent studies. Uttarakhand is the hilly state of India and the most vulnerable to landslides. Some studies have been done to predict landslide in Uttarakhand using ML techniques, but there is scope for more studies to predict landslides in the Uttarakhand prone region using recent ML techniques such as CNN.

Keywords: Machine learning, LSM, hybrid, RF, SVM, ANN, CNN, ROC, AUC.

I. INTRODUCTION

Every year thousands of humans lose their life in various natural disasters like flood, fire, cloud burst, earthquake and landslides. In all over the world the frequency of natural disasters like landslides are increasing in recent past and it causes harm to human life. And in near future, it is likely that more landslides will happen, due to various conditioning factors. Landslide is a natural disaster that mostly occurs in hilly regions. Landslide affects the whole world and it is a big cause of economical loss, infrastructure loss and it also puts human life in danger. Many climatic and geographical factors cause landslide to occur. Out of these factors, the most influencing is heavy rainfall, a large amount of cutting trees in the hilly region, and earthquakes. Due to landslide, a long-lasting bad impact occurred on the environmental and economic state.

II. LANDSLIDES IN UTTARAKHAND STATE

The region of the Himalayas is home to approximately 80% of tribulations in India. In northern India, Uttarakhand is a province. The natural environment in the Himalayas is renowned for Uttarakhand. The Uttarakhand mountains are geologically extremely well positioned and the climate changes rapidly and makes them the most tributary area in Indian territory. Some 0.42 million sq. in India. The landslide risk of this 0.14 million sq km or 12.6%, with the exclusion of a snow-capped area. The Himalayas (Uttarakhand, Himachal and Jammu & Kashmir) are situated in northwestern North Himalayas. The Himalayan terrain is also one of the highest areas that are vulnerable to an earthquake (Zone-IV and BIS 2002) where earthquakes with the strength of Modify Mercalli VIII-IX can occur and are thus prone to earthquake-caused terrain disappearances. Studies showed that during the period from 2004 to 2010, there were 2620 fatal non-seismic landslides occurred in the world, in which 32,322 people lost their lives in all over the world. From the year 2004 to 2016, about 55997 deaths have been reported due to 4862 landslides. The Bhabar and the Terai regions. Uttarakhand (India) is the highly prone area for natural disasters including landslides, earthquake, forest fire and floods. In which, the landslide is the most frequent disaster. Various places in Uttarakhand are sensitive for landslide where several landslide events occur every year. Pithoragarh district, Uttarkashi, Chamoli, Pauri and Tehri, are the landslide-prone area of Uttarakhand. Every year due to landslides, the Uttarakhand state has to bear a loss of lives, infrastructure and ecological environment. To avoid this destruction, it is therefore effective to recognize areas that are landslide-prone.

III. LANDSLIDE SUSCEPTIBILITY MAPPING (LSM)

Landslide vulnerability map is used as an efficient pre-planning and decision-making method for preventing loss due to landslide in areas that are vulnerable to slipways. The most vulnerable region of the landslide in Uttarakhand, a hilly state, so the danger of life and property is still present. LSM for Uttarakhand has become relevant. Landslide susceptibility map

reveals landslide susceptibility areas and forecasts potential landslides. LSM recognizes various landslides phenomena and different GIS modelling efficiently. It is also helpful in disaster recovery management by collecting relevant data, recent landslide event history, and risk analysis of landslides future events. The GIS techniques is very helpful in data collection, data monitoring and data mapping of susceptible landslides. The better utilization of land data and efficient risk analysis of landslides on earth surfaces data through GIS is very essential for LSM.

IV. MACHINE LEARNING: AN EFFECTIVE TOOL FOR LSM

Machine learning is an artificial intelligence subset that can manage big data. It can be used to build new technology to offer information more accurately. These models based on machine learning can predict disaster and allow disaster response teams to respond. ML models can produce information that could be useful in decision-making by analyzing vast data from past events. It is possible to predict potential landslides with the use of ML models. ML is an artificial intelligence, learns without being specifically programmed, and enhances output automatically with past experiences. We have input and output data when we learn machine; with this information, our machine is trained and our model or software is provided as an output. New input data can be used to validate this software. By extensive review of many previous kinds of research, it is found that supervised ML algorithms can be explored to develop models that can produce landslide slide susceptibility maps. Using the landslide susceptibility map, we can prevent human casualties and economical loss. To predict landslides, several researchers used various ML approaches. Some of them used

LSM CNN. Some have used the CNN. CNN is an evolving method of a deep education in this LSM field. It has been documented. In this analysis, we believe that the CNN method can be implemented in the Uttarakhand area to forecast LSM. Our goals are to establish a methodology of deep learning (CNN) by measuring the key factors that are affecting the development and validation of a CNN-based LSM model. Fig. 1 offers a description of the ML technology applied to LSM.

V. LANDSLIDE PREDICTION WITH VARIOUS MACHINE LEARNING MODELS

In the past years by applying ML methods many researchers developed ML-based models for LSM. Different machine learning models as a logistic model tree (LMT), random forest (RF), classification and regression tree (CART) and many more were applied for LSM and they showed better results in comparison with statistical methods. In recent years many machine learning models were compared with each other to find the best for LSM, like Least Square Support Vector Machine (LSSVM), Hodrick-Prescott decomposition, Double Exponential Smoothing (DES) method, Proximal Support Vector Machine (PSVM), L2-Support Vector Machine-Modified Finite Newton (L2-SVM-MFN), Maximum Entropy (MaxEnt), Support vector machine (SVM) and Artificial Neural Network (ANN) and their different combinations (ANN-SVM, ANN-MaxEnt, ANN-MaxEnt-SVM, SVM-MaxEnt) were compared to predict LSM. SVM have gained highest spatial area is highly susceptible classes and ANN gained the improved curve (82.4%) AUC individually while ensemble learning ANN with SVM gained highest AUC (82.6%) out of all ensemble models [1]. RF, Gradient Boosting Machine (GBM), Logistic Regression (LR), Artificial Neural Network (NNET), and SVM

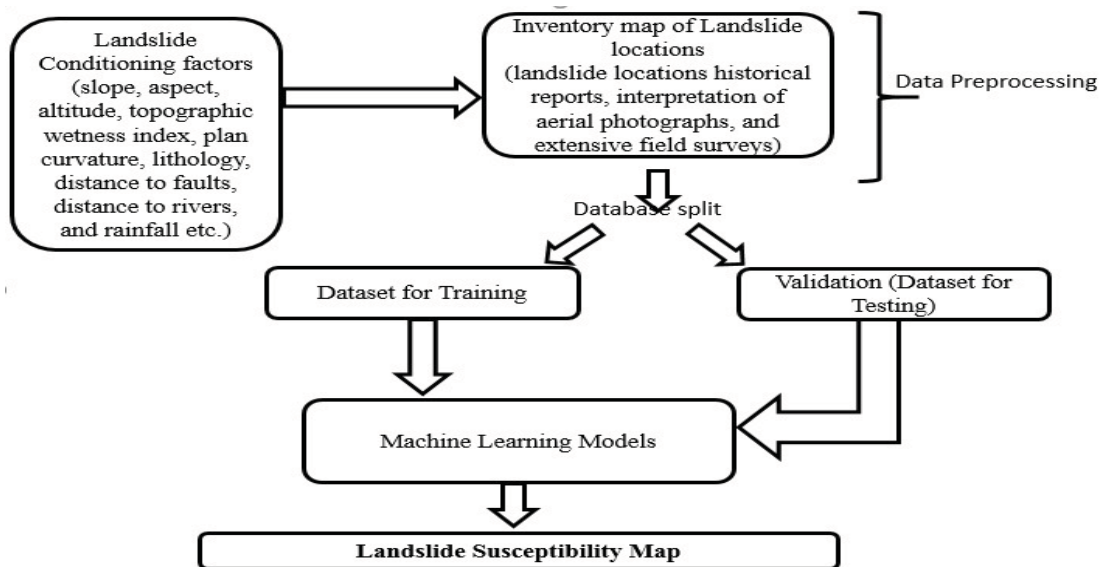


Fig. 1. An overview to produce LSM by ML methods used previously in different studies.

ML techniques were used to build landslide susceptibility models. In their database, 1156 landslide polygons and 16 conditioning factors were used. They split the database into training and testing data and applied 10 k fold cross-validation. They found that GBM model achieved the highest prediction capability ($AUC = 0.8967$) in comparison of other models. RF secured the second position by scoring $AUC = 0.8957$. They conclude that GBM and RF are the most suitable ML techniques to build land susceptibility prediction models [2]. A correlative selection function (CFS), RF, and a colony optimization (ACO) were used in a study feature selection and classification model. To demonstrate that this data enhances the difference between shallow and deep-sitting slides, laser transmitters and receivers have been used to detect and range (LiDAR) data. For the extraction of the optimizer parameter, the Fuzzy dependent segmentation parameter (FbSP optimizer). The findings have shown that RF and CFS are good for distinguishing landslide forms [3].

It is shown that SVM improves when SVM for the prediction and classification of landslide susceptibility groups is being applied for the optimization of particle swarm. This model classification is very large, high, moderate and small on the landslide susceptibility index. Database constructing was used for remote sensing and geographical information technology. The results showed that PSO-SVM and genetic programming performed well with achieved precision = 94.5%, and the genetic programming procedure also worked well with almost 100% accuracy. Geographical information systems (GIS) were compared and found that RF performed better than SVM, NB, RF, $AUC=83.3$, $SVMAUC=75.7$ and $NB AUC=72.5$. The computer learned three models, namely SVM, RF and NB. For this analysis, 146 landslide sites and landslide inventory maps were identified from field surveys and 12 conditioning factors were interpreted. For training and testing data set, the full data set was divided into 70:30. Relation has been identified between landslide conditioning and landslide frequency [6].

A study was done to create gully erosion susceptibility map (GESM) using four machine learning models namely RF, NB, SVM and generalized additive model (GAM) also revealed that RF model achieved highest predictive accuracy $AUC=92.4\%$ as compared with SVM, GAM and NB whose AUC score are 90.9%, 89.9% and 87.2% respectively. To identify the spatial distribution of soil surface 130 locations were sampled. Using digital elevation model (DEM), Topographic attributes were prepared. In 2020, Bayesian optimization algorithms were used by the RF model to optimize the hyperparameters. 16 landslide conditioning factors and 1520 historical landslides were selected and geospatial data were developed. The method of recursive function elimination has been used to remove the most promising conditional factors. The result showed that RF after hyperparameter optimization, model achieved $AUC=0.95$ on train data set [5]. Some of the

researchers also checked the accuracy of Artificial Neural Network (ANN) for prediction of LSM ANN performs like a human brain when there is huge imbalance data. But when the class has very few data samples ANN prediction performance degraded. According to the environmental states, a study change strategy was suggested which can choose from different predictors. This proposed method improves the prediction capability of ANN [6]. The ANN-based model achieved 82.5% accuracy when Mathematical criteria, sensitivity analysis, previous studies and actual landslide location were used to identify the main influencing factors for the study. 14 landslide conditioning factors were prepared and a landslide inventory map from 242 events was identified after interpreted photographs to generate high-resolution landslide maps [7]. In 2019 a new approach focused on LSM in which Convolutional Neural Network (CNN) model was proposed. CNN is a kind of artificial neural feed network where the pattern of communication between the neurotic cortex is influenced by the organization of the animal visual cortex, and a neuron in the layer is only linked in CNN to a small layer region before it. Researchers used CNN to compare it to LSM and other ML models as well. For the first time CNN was used for LSM in Yanshan country, the region of China by Y. wang et al 2019 and it was compared with SVM and Deep Neural Network (DNN). In this study 16 landslide conditioning factors were involved and historical landslide location was divided 70:30 ratios for training and testing. The result showed that CNN achieved a good AUC score for landslide susceptibility mapping. Proposed CNN architecture is better than SVM and Deep Neural Network (DNN) [8]. CNN also compared with SVM and ANN for LSM in Southern Yangyang Province South Korea. One dimensional Neural Network (1D-CNN) showed better accuracy ($AUROC=0.893$) than SVM ($AUROC=0.807$) and ANN ($AUROC=0.79$). For the study, 219 Historical landslide locations and 17 landslide conditioning factors were prepared and Bayesian optimization was used to select significant CNN hyperparameters [9]. Dao et al in 2020 also compared Deep learning (DL) neural network with other machine learning models for LSM. A disadvantage of this model is it required high computation power when the database is large. For this study 217 landslide event in Nuong Lay district, Vietnam and 9 landslide conditioning factors were used to construct the database. DL model performs better in training ($AUC=0.90$) and a testing ($AUC=0.89$) [10].

VI. LSM PREDICTION BY HYBRID ML MODELS

Many researchers proposed Hybrid models for LSM. Hybrid models are combinations of different ML models. More than one machine learning techniques are combined to generate more accurate prediction is called ensemble machine learning. BAGging or Bootstrap AGGregating are some of the types of

ensemble learning. Random Forest is also an example of ensemble machine learning. So in this section, we are discussing hybrid machine learning models that were used by researchers for LSM. In most of the cases, SVM combined with other models to produce more accurate LSM. For many uncompromising issues, including dimensionality, limited known samples, and noise corruption, kernel-based classification is significant. SVM-based models are used in many LSM studies. The SVM's downside is that the SVM-based model is labelling uncertainty; they consist of the attribute value of any conditioning factor and neglect the adjacent vector's high dependence with valued grid cells. The combination of influential factor, attribute factors with spatial addition can overcome this problem. SVM has been combined with various LSM composite kernels for analysis purpose. This research considered 16 landslide situations and 364 historical landslides. This study introduced two types of composite kernel: first the single kernel with stacked vector and second, the summing kernel. The key benefit of the proposed framework was the use of information integration to improve the efficiency and the extraction of useful information from the spatial domain. ROC AUC = 0.8900 is 0.0111 higher than the original SVM in a summation kernel SVM. One of the main factors for the displacement of landslides is heavy precipitation. Much of the time the rainy seasons lead to a landslide. In China, work has recently been carried out to predict the vulnerability to landslides through ensemble learning. CNN is a category of SVM, RF, and LR in this analysis. Seventeen conditions for slide modelling have been established. The findings showed that CNN is the best way to derive features and CNN when ensemble with SVM, RF, and LR have better predictions than individual classifiers [18]. 364 historical landslides have been divided into 70:30 ratios. For assessing the vulnerability of landslides in the Pithoragarh, Uttarakhand, The Himalayas, India a hybrid machine learning algorithm was applied. In this process, the combination of an alternating decision tree (AD tree), three different ML models, bagging (BA), random subspace (RS) and rotational forest (RF) were used. RS-ADTree model (AUC training = 0.883) and RF-AD Tree (AUC training = 0.871; AUC validation = 0.840) model outperforms the AUC model. RS-ADTree model (AUC training = 0.883; AUC training = 0.842). Therefore, the RS-AD Tree ensemble model demonstrated maximum accuracy so that it can be used as a modern technique for the prediction of slide-prone areas. After the evaluation of the chi-square attribute, this analysis also revealed that the ground cover was observed as the principal factor and landslide that did not occur with curvature from various significant factors [11]. Three data-mining techniques were used for LSM in Hanyuan County (China), including Adaptive Neuro-Fuzzy Inference Method (ANFIS) combined with the Frequency Ratio (ANFIS-FR), the Generalized Additional Model (GAM) and the Supporting Vector Machine.

12 conditioning factors have been taken into account for this analysis. The application of the collinearity test and correlation analysis established significant conditioning factors. Educated and tested were ANFIS-FR, GAM, and SVM. ANFIS-FR achieved 0.875, GAM 0.846 [4], and ANFIS-FR 0.851 obtained the highest prediction score. In the same year, a space prediction system model for shallow terrain was proposed. In this research, a hybrid machine learning model called BALSSVC was created using Least Square Support Vector Classification (LSSVC) and the BAT (BA) algorithm. The Long Son (Vietnam) Geographic Information System was used to develop databases. During the training stage of LSSVC BA metaheuristic was used for LSSVC to assist in the above classes. The LSSVC classes were categorized as non-landlines and landslides. LSSVC was used to divide the data into two classes. The results show that BALSSVC achieved 90% exactness in the spatial prediction of shallow terrain and did better than CNN Logistic regression [12]. The Landslide Hazardous Indicator (LHI) was manufactured by Stanely et al in 2020 to estimate possible landing times and locations. Data from the national climate assessment were obtained in this analysis of 36 years. To predict the frequency and likelihood of land-life, geographical data, and climatic data were used. XGBoost software was used to predict landslides with the Gradient boosting tool. This research has therefore recently revealed that LHI can be used to predict, compare, and guide the seasonality of sliding on the land [13]. For landslide susceptibility, spatial heterogeneity was considered. Spatial heterogeneity is the relation between additive landslide and individual landslides. The study region was grouped into multiple clusters by GeoSOM. The same type of susceptibility factors is associated with each cluster. Heterogeneity has been overcome in this way by GeoSOM. A high-level landslide model was designed by integrating 3 conventional SVM, ANN, and GBDT methods of machine learning. The stacking ensemble technique was used to develop a high-performance gradient model. The proposed model worked better than other models of machine learning. In this study, GeoSOM has also been used to solve this problem [14]. A study found that both deep learning and ML combined classifying systems provided good predictive results than just a single classification for the assessment of landslide susceptibility. The terrestrial evaluation was carried out using auto encoding in conjunction with random forests. The result was rainfall. A stacked Auto-encoder (StAE) and a sparse SpAE (SpAE) are composed of a raw-data input layer, a hidden feature extraction layer, and a graded and projected output layer. RF showed 93.2% (AUC) prediction ratio in combination with StAE and 92.5% (SpAE) in combination. The prediction ratios were higher than SVM ratios (87.4%), RF (89.7%), StAE ratios (84.0%), SpAE ratios (88.2%) [15]. ANN ensemble (ANNE) was used to map the vulnerability of landslides in a recent study L. Bragagnolo et al 2020. The study employed historic landslide

locations and seven parameters, such as height, appearance, slope, topographic humidity index, curvature profile, lithology, and usage of the ground [16]. The result shows that in the

landslide susceptibility mapping, ANNE has improved its precision [Table 1].

Table 1: Progression in LSM using different ML models from the Year 2017 onwards.

Year	Authors	Models	Result
2017	W. Chen et al. [17]	MaxEnt ,SVM , ANN and Ensemble models (ANN with SVM, ANN with MaxEnt, ANN with MaxEnt and SVM,SVM with MAXEnt)	ANN with SVM gained the highest AUC (82.6%) out of all ensemble models.
2018	Motevalli et al. [6]	SVM, RF and NB	The result showed that RF performed better than SVM and NB, RF achieved AUC=83.3, SVM AUC=75.7 and NB AUC=72.5
2019	B.T. Pham et al. [11]	BREPT, MBREPT, RFREPT, RSREPT	BREPT observed best in the prediction of LSM
2019	D. Utomo. [6]	Switching strategy-based model to choose between different predictors, based on environmental factors for ANN	This proposed method improves the prediction ability of ANN
2019	D. T. Bui, et al. [13]	LSSVC and BA	BALSSVC achieved 90% accuracy in spatial prediction of shallow landslide and performed better than CNN, Logistic regression.
2019	Y. Wang et al. [8]	Proposed CNN for Landslide susceptibility and compared with some other ML methods SVM and DNN	CNN achieved better AUC=0.813 than SVM and Deep Neural Network
2019	A. A. Shahri et al. [7]	ANN-based model for landslide susceptibility model	ANN base model achieved 82.5%.
2020	Y. Wang, et al. [14]	Hybrid Machine learning model with SVM, ANN and GBDT	This study concludes that spatial heterogeneity also affects to generate landslide susceptibility by ML models so in this study GeoSOM was used to overcome this problem.
2020	D.V. Dao, et al. [10]	Deep learning neural network	AUC=0.89
2020	L. Bragagnolo et al. [16]	Artificial neural network ensembles (ANNE)	The result showed that ANNE achieved higher accuracy in landslide susceptibility mapping.
2020	M. I. Sameen et al. [9]	Proposed 1D-CNN, with Bayesian optimization to predict landslide susceptibility mapping.	The result showed that 1D-CNN achieved better accuracy (AUROC=0.893) than SVM (AUROC=0.807) and ANN (AUROC=0.79)

VII. CONCLUSION

Landslide is a natural hazard that leads to the loss of life and land every year. We will minimize the loss of life and property if the landslide is expected. The study concludes that the use of ML technologies can predict disappearances. Studies have shown that various ML techniques and their different combinations are used to predict LSM. It has been stated that SVM is a technology that compares the outputs of other ML models for LSM and still for SVM. Some papers have reported the use of other ML techniques, such as RF and ANN. Many hybrid techniques (ensembles) for LSM, better than the use of a single ML technique for LSM, were also approached. CNN has also been focused on the forecast of earthquakes in recent years. Based on previous studies, we assumed that it would also be helpful to discuss CNN for LSM in Uttarakhand.

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